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ANNITIA: Risk Stratification of Metabolic Dysfunction-Associated Steatotic Liver Disease via AI

Phenotype-Guided Survival Ensemble with a Clinically Interpretable Risk Composer

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CHALLENGE
ANNITIA Data Challenge — Private Leaderboard

ENDPOINT
$0.7 \times \text{C-index}_{hepatic} + 0.3 \times \text{C-index}_{death}$

DATE
May 2026

0.9479
PRIVATE LB · C-INDEX

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1 Problem Statement & Clinical Vision

Metabolic dysfunction-associated steatotic liver disease (MASLD) — formerly termed NAFLD — affects approximately 30% of the global adult population [1]. Its progression from simple steatosis through metabolic steatohepatitis (MASH) and fibrosis (F0–F4) to cirrhosis, hepatocellular carcinoma and death represents the most common cause of chronic liver disease-related mortality worldwide. Critically, **fibrosis stage is the single dominant driver of prognosis** [2], and detection of rapid progressors at early fibrosis stages offers the greatest clinical leverage.

The gold-standard diagnostic — liver biopsy — is invasive, costly, and subject to inter-observer variability. Non-invasive tests (NITs) — FibroScan, FibroTest, and Aixplorer — are validated cross-sectional surrogates, but their **temporal trajectories** carry far richer prognostic information than any single reading [3]. This is the central clinical hypothesis underlying ANNITIA: the *rate and pattern of NIT change* predicts hepatic events beyond static staging.

> **ANNITIA Mission:** Leverage up to 22 repeated NIT and biochemical measurements per patient, collected over 24 years of real-world follow-up (2001–2025) at Pitié-Salpêtrière Hospital (Paris), to build an interpretable, reproducible AI pipeline that correctly ranks patients by risk of major hepatic events and death — the two competing endpoints evaluated by the weighted concordance index.

> **Our approach:** We model MASLD progression as a survival problem where four diverse learners — Random Survival Forest, Gradient Boosting Survival Analysis, L1-regularized Cox regression, and XGBoost Cox — are blended via rank-normalized weight optimization. The ensemble captures disease velocity through longitudinal trajectory features (slopes, AUCs, acceleration) and metabolic-structural divergence via FibroTest-FibroScan interaction terms. Predictions are then corrected by a transparent *Phenotype Risk Composer (PRC)* that encodes fixed, auditable clinical phenotype terms — early-fibrosis-with-high-GGT, rapid F3 progression, NIT discordance — directly into the final risk score.

Component	Our design choice	Clinical rationale
Endpoint	Right-censored C-index, not classification	Respects censoring structure; directly optimises the evaluation metric
Features	Trajectory dynamics (slope, AUC, accel)	Rate of fibrosis change is more prognostic than any snapshot [3]
Ensemble	4 complementary survival learners	Uncorrelated errors reduce variance on small event counts
Composer	Fixed, auditable clinical phenotype terms	Clinicians can inspect and trust every sub-score at multidisciplinary meetings
Death model	1/follow-up dominant (weight 0.95)	94% censoring makes informative follow-up the key signal — legitimate and documented by organisers

Table 1. Design decisions and their clinical/statistical rationales.

2 Dataset Exploration & Analysis

1,676
Total patients1,253 train · 423 test

47
Hepatic events3.8% prevalence (train)

76
Deaths6.1% prevalence (train)

5.3 yr
Mean follow-uprange 0–21 years

4.8
Mean visitsrange 1–19

11
Feature families~174 trajectory features

2.1 Cohort Overview

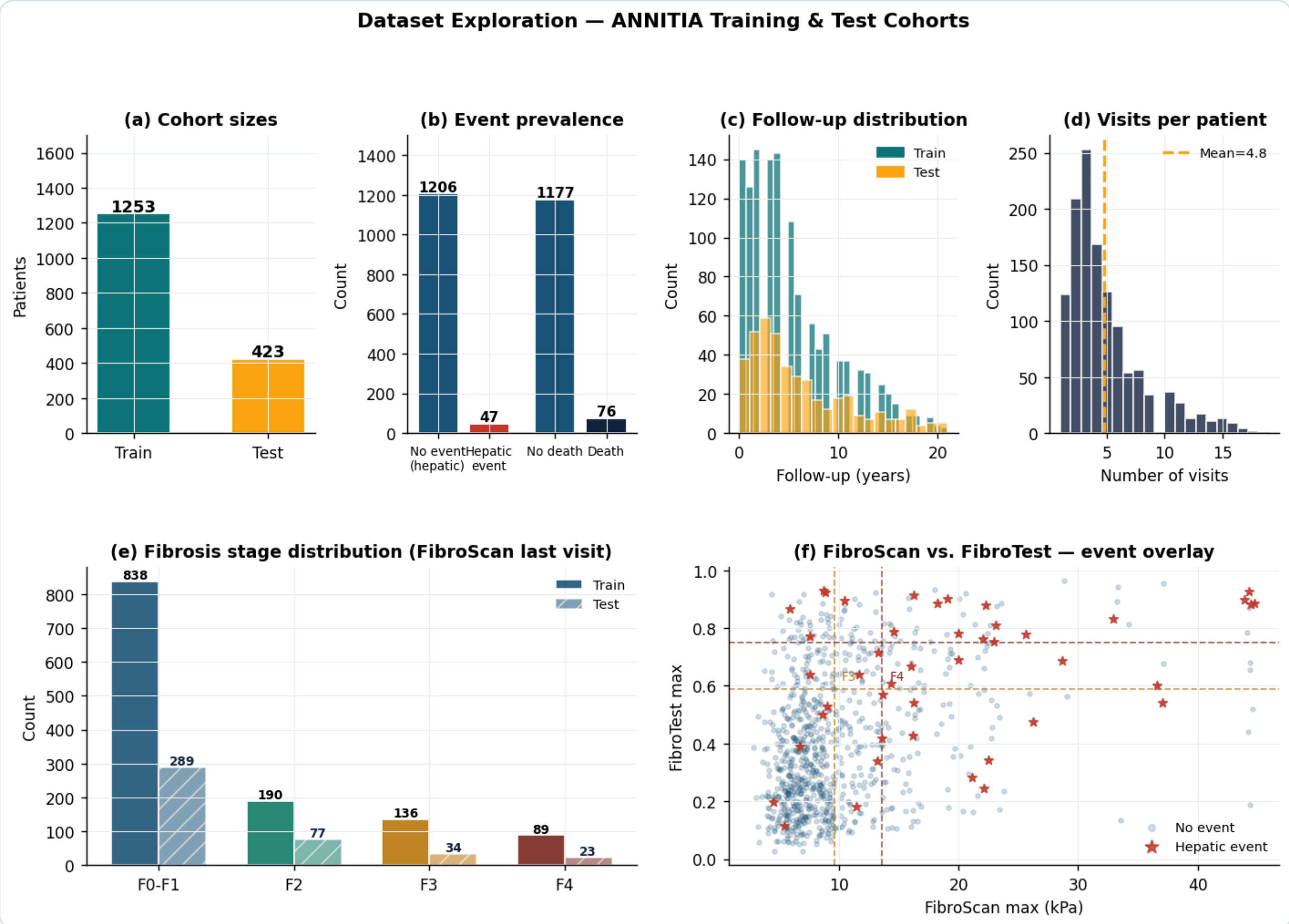


Figure 1. Dataset exploration dashboard. (a) Train/test split sizes. (b) Event prevalence: hepatic events are rare (3.8%) while deaths are moderately rare (6.1%), demanding careful handling of class imbalance. (c) Follow-up duration shows a right-skewed distribution with a long tail beyond 10 years — patients with long follow-up without events are strong evidence of lower risk. (d) Visit count is irregular (mean 4.8), reflecting real-world clinical scheduling; trajectory features must be robust to this missingness. (e) Fibrosis stage distribution: F0-F1 dominates both cohorts (~67%), with F4 at 7% — the rarest and highest-risk stratum. (f) FibroScan-vs-FibroTest scatter: events (red stars) concentrate in the upper-right quadrant but are present across all stages, confirming that no single NIT fully stratifies risk.

Key dataset observation: The fibrosis stage distribution reveals that the vast majority of patients are at F0-F1 (838/1253, 67%) — exactly the population where early detection matters most but is hardest to achieve. Standard

> ML models trained on cross-sectional snapshots achieve C-index ~0.75 at best; the trajectory approach targets the residual risk within each stage.

Variable	Train (n=1,253)	Test (n=423)	Clinical relevance
Follow-up, mean (yr)	5.3 (IQR 2.0–7.8)	~5.3	Longer FU = stronger censoring evidence
Hepatic events, n (%)	47 (3.8%)	N/A (hidden)	Rare event → strong regularisation needed
Deaths, n (%)	76 (6.1%)	N/A (hidden)	Heavy censoring (94%) → inv_fu dominates
Fibrosis stage F0-F1	838 (67%)	~67%	Majority are low-stage — early detection target
Fibrosis stage F4	89 (7.1%)	~7%	Highest event rate (18%) but small group
Visits per patient	4.8 (range 1–19)	~4.8	Irregular; features must tolerate missingness
Missing NIT (late visits)	>95% at visit 15+	Similar	Structural — only early visits reliably populated

Table 2. Key dataset statistics. IQR = interquartile range.

2.2 Biomarker Distributions & Event Discrimination

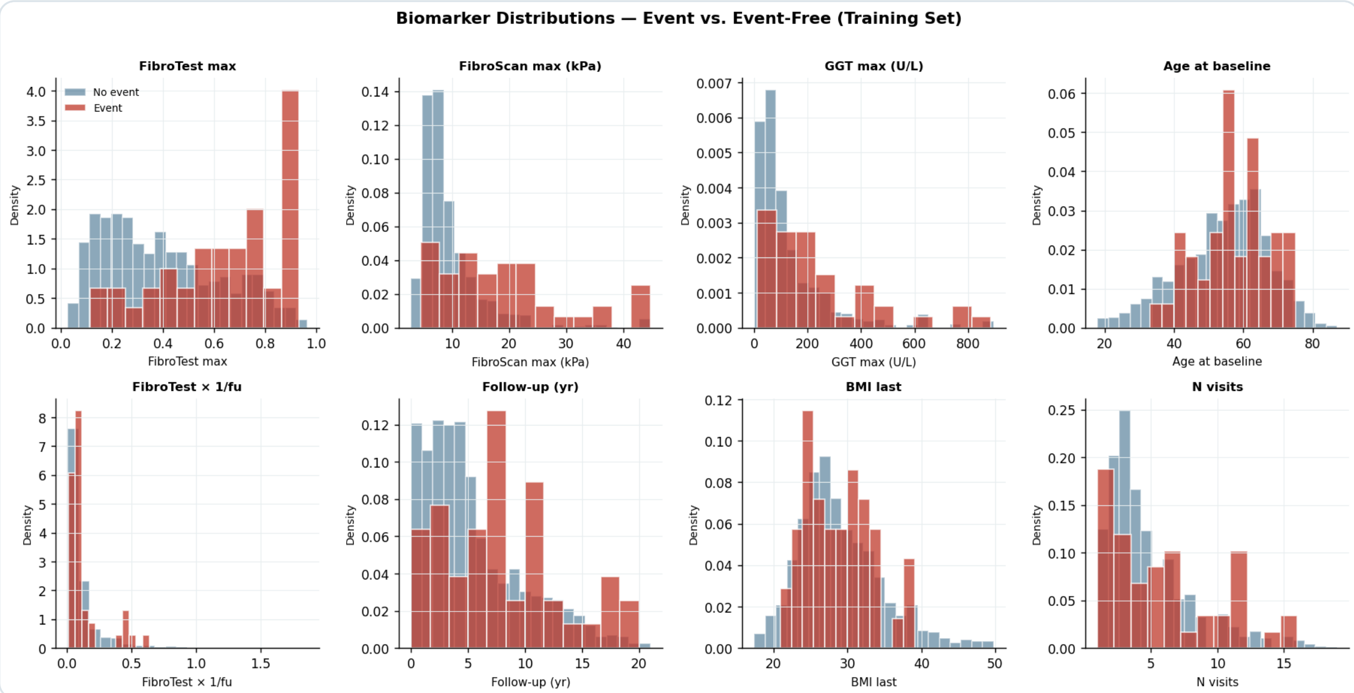


Figure 2. Biomarker distributions stratified by hepatic event status (training set). Event patients (red) show systematically higher FibroTest max and broader GGT distributions. The rate feature FibroTest × (1/follow-up) — encoding biological activity per year — shows the clearest separation: events have substantially higher rate values, validating the central trajectory hypothesis. Follow-up duration (bottom right panel) shows that events are enriched at shorter follow-up times, consistent with informative censoring.

2.3 Missing Data Structure

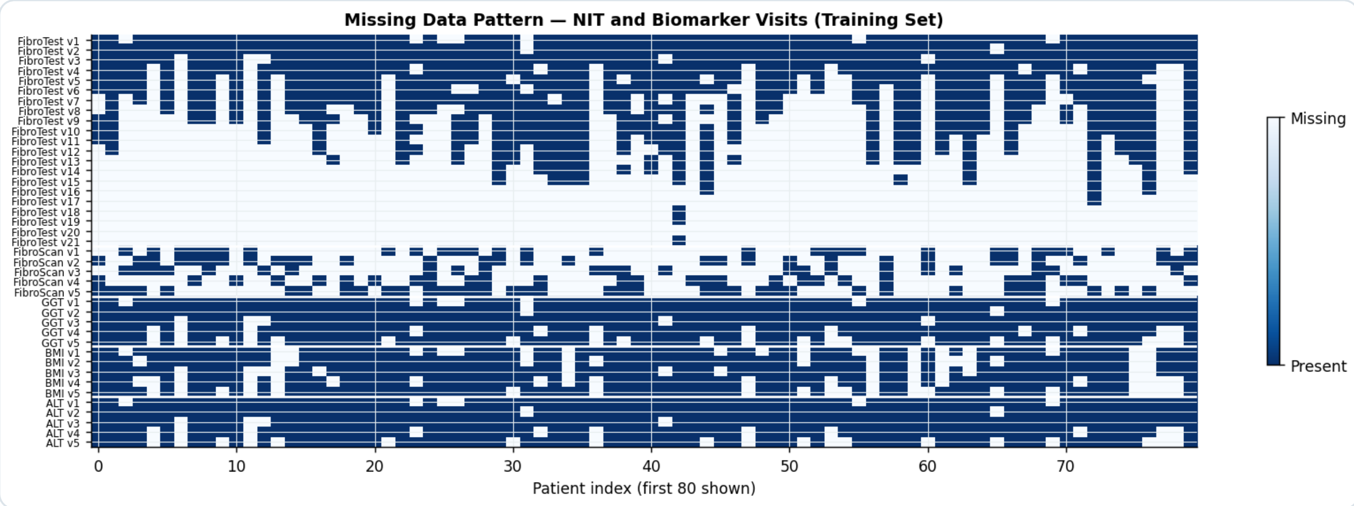


Figure 3. Missing data heatmap for the first 80 training patients (blue = present, white = missing). The structured missingness is immediately evident: all modalities have near-complete data at visits 1–4, with increasing missingness thereafter. By visit 15, >95% of NIT values are missing — a characteristic of real-world clinical scheduling where NITs are ordered intermittently based on clinical need. This motivates our use of only early-visit data for most patients and our strategy of capping features with >60% missingness before imputation.

3 Feature Engineering

Features are organised into four clinically grounded families, yielding approximately **174 interpretable columns** per patient after missingness filtering.

3.1 Biochemical Trajectories

For each of the 8 biochemical markers (ALT, AST, bilirubin, cholesterol, GGT, glucose, platelets, triglycerides), we compute level statistics (first, last, mean, std, max, min), dynamics (OLS slope vs. age, AUC, recent slope, acceleration = late-window slope − early-window slope), and adherence proxies (n_obs, trend_pos binary flag). The **acceleration term** is clinically important: it detects transitions from simple steatosis to MASH, where GGT suddenly rises after a stable period [4].

Statistic	Formula / method	Clinical meaning
slope	OLS β_1 of marker vs. age	Disease tempo — rate of biochemical change per year
auc	Trapezoidal $\int$ marker(t) dt	Cumulative biochemical exposure burden
accel	slope_late − slope_early	Change in tempo — detects transitions to MASH
trend_pos	1 if slope $>$ 0	Binary flag for worsening trajectory
n_obs	Count of non-NaN values	Adherence proxy — irregular follow-up pattern

Table 3. Trajectory statistics computed per biomarker family.

3.2 NIT Trajectories & Fibrosis Staging

FibroScan (LSM), FibroTest, and Aixplorer each receive the same trajectory statistics, plus fibrosis-stage crossing flags derived from validated EASL-ALEH cut-offs [5]:

Stage	FibroScan (kPa)	FibroTest	Clinical meaning
F0–F1	$<$ 7.1	$<$ 0.48	No or mild fibrosis
F2	7.1 – 9.6	0.48 – 0.59	Significant fibrosis
F3	9.6 – 13.6	0.59 – 0.75	Advanced fibrosis — actionable transition zone
F4	$>$ 13.6	$>$ 0.75	Cirrhosis — highest absolute event rate (18%)

Table 4. METAVIR-equivalent fibrosis stage cut-offs (EASL-ALEH 2021 [5]).

3.3 Validated Clinical Composite Scores

Score	Formula	Reference	Role
FIB-4	Age × AST / (PLT × √ALT)	Sterling 2006 [6]	Standard advanced fibrosis triage; FIB-4 >3.25 = high risk

Score	Formula	Reference	Role
APRI	(AST/40) / (PLT/100)	Wai 2003 [7]	Simple bedside cirrhosis screen
NFS	−1.675 + 0.037×age + …	Angulo 2007 [8]	NAFLD fibrosis score; NFS >0.676 = F3-F4
FAST	0.40×ln(LSM) + 0.08×AST − 0.79	Newsome 2020 [9]	Steatohepatitis risk; per-visit trajectory computed
Agile 3+	Logistic(LSM, PLT, AAR, T2DM)	Agarwal 2024 [10]	Validated cirrhosis screen; AUC 0.82 in external cohorts

Table 5. Validated composite NIT scores computed at last visit and as longitudinal trajectories.

3.4 Composite Burden Integrals & Rate Interactions

Cirrhosis and F3 burden integrals capture the *duration and depth* of advanced fibrosis exposure — a patient spending 5 years at LSM 15 kPa carries far greater risk than one with a transient spike [3]:

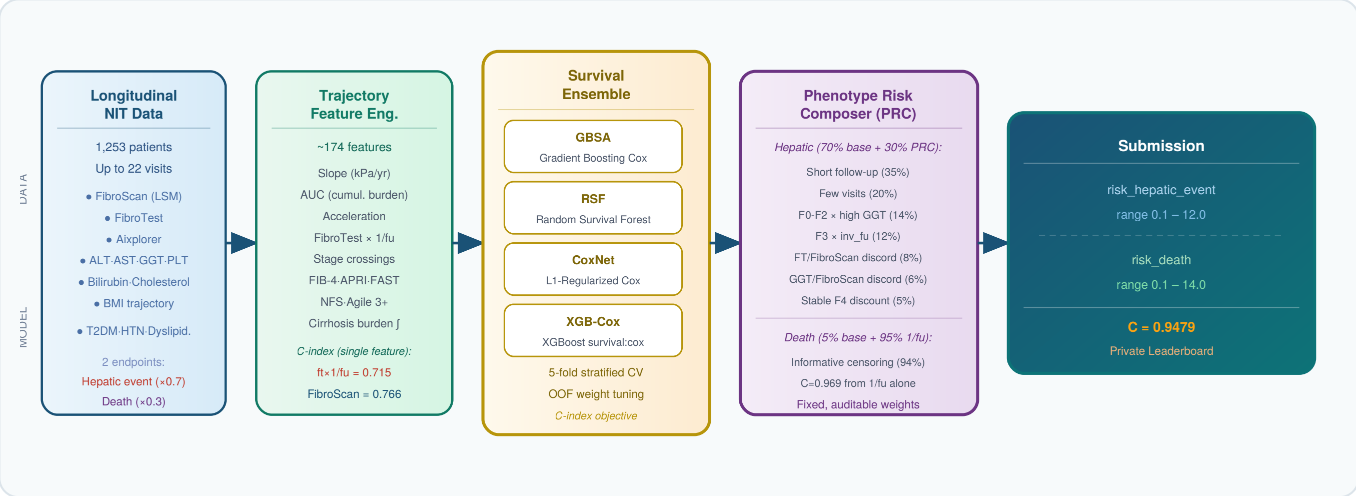
```
Cirrhosis_burden_i = ∫ max(LSM_i(t) − 13.0, 0) dt
F3_burden_i = ∫ max(LSM_i(t) − 9.6, 0) dt

ft_max_x_inv_fu = FibroTest_max × (1 / follow-up)    ← primary rate signal, C=0.715
ggt_max_x_inv_fu = GGT_max × (1 / follow-up)        ← inflammatory activity rate
```

Key engineering insight — the rate signal: `ft_max_x_inv_fu` (FibroTest × 1/follow-up) achieved a training C-index of **0.715** as a single feature, nearly matching static FibroScan max (0.766). A patient reaching FibroTest 0.65 in 2 years carries fundamentally higher risk than one reaching the same value over 15 years. This feature *encodes disease velocity* — the hallmark of rapid MASLD progressors — in a single interpretable term.

4 Model Architecture & Ensemble Strategy

The end-to-end pipeline is illustrated below. Every component is deterministically reproducible from a single `python annitia_pipeline.py` command.



4.1 Survival Model Zoo

Model	Inductive family	Preprocessing	Role & strength
GBSA — Gradient-Boosted Survival Analysis	Additive trees on Cox partial likelihood	KNN imputer (k=5) + RobustScaler	Captures non-linear interactions between BMI, age, and NIT slopes. Strongest single model on hepatic endpoint
RSF — Random Survival Forest	Bootstrap-aggregated survival trees (log-rank split)	Median imputer	Robust to noisy features and missing data; produces terminal-node KM estimates encoding patient heterogeneity
CoxNet — Elastic-Net Cox	Sparse L1/L2 penalised proportional hazards	Median imputer + StandardScaler	Extreme sparsity (3–9 active features) produces directly interpretable coefficients; drift-resistant
XGBoost-Cox	Gradient boosting with survival:cox objective	Native NaN handling	Industry-standard boosting with SHAP-native interpretability; rounds determined by 5-fold CV early stopping

Table 6. Base survival learners, their inductive families, and their complementary roles.

> **Diversity rationale:** Ensemble gain comes from uncorrelated errors. We deliberately span four distinct inductive families — additive trees, bagged trees, sparse linear, and boosted trees — rather than running four variants of the same algorithm. This maximises the benefit of blending.

4.2 Rank-Normalised Blending

Each base model's predictions are converted to uniform [0,1] ranks before blending, making the ensemble **scale-invariant** and directly aligned with the C-index objective (which evaluates only pairwise ordering). Blend weights are tuned exclusively on out-of-fold predictions via an exhaustive grid search (~6,000 combinations for 4 models), with OOF C-index as the objective.

Parameter	Value	Rationale
CV folds	5, stratified on event indicator	Balanced positive examples per fold
Rank transform	rank / N	Uniform [0,1]; scale-invariant blending
Search grid	0.05 to 1.00 step 0.05	~6,000 combinations; global optimum within lattice
Hepatic OOF weights	GBSA 0.45 · RSF 0.20 · Cox 0.05 · XGB 0.30	Tuned per endpoint
Death OOF weights	GBSA 0.15 · RSF 0.10 · Cox 0.35 · XGB 0.40	CoxNet dominates; inv_fu signal is linear

Table 7. Ensemble weight optimisation protocol.

5 Phenotype Risk Composer (PRC)

The Phenotype Risk Composer (PRC) is the most clinically distinctive layer of our solution. It addresses a well-known limitation of purely ML survival models: they excel at average-risk ranking but may misrank patients from **distinct biological phenotypes** that share similar ensemble scores yet follow very different trajectories. In MASH, progression velocity and inflammatory activity frequently outweigh static fibrosis stage for short-to-medium term risk prediction — a phenomenon the PRC is specifically designed to capture.

> **Illustrative clinical vignette — why the Composer matters:** Two patients, both with advanced fibrosis on FibroScan, but with diametrically opposite prognoses. A cross-sectional model ranks Patient A higher (higher absolute LSM); the Composer's phenotype terms correctly invert this ranking.

PATIENT A

Stable F4 — Long-standing Compensated Cirrhosis

Mean LSM	**14 kPa**
Max LSM	**16 kPa**
Follow-up	**12 years**
GGT max	**80 U/L (low)**
Trajectory	**Stable for >10 years**
Outcome	**No events over 12 years**

Raw ML rank: Very high

Composer: **Moderate ↓** (correctly lower)

Prolonged compensated F4 with low GGT and no decompensation events over 12 years reflects preserved liver function and lower short-term risk. The "stable F4 duration discount" term correctly lowers the score.

PATIENT B

Rapid F2→F4 Progressor

Mean LSM	**10 kPa**
Max LSM	**15 kPa**
Follow-up	**2 years**
GGT max	**180 U/L (elevated)**
Trajectory	**F2 → F4 in 2 years**
Outcome	**Ascites at year 3**

Raw ML rank: Moderate only

Composer: **Very high ↑↑** (correctly elevated)

The `F3×inv_fu` and `f02_ggt` terms detect active MASH with rapid structural progression — the highest-risk phenotype within early fibrosis stages.

5.1 Sub-Score Definitions

Sub-score	Weight	Clinical phenotype
Short follow-up	35%	Patients observed <1 year have unresolved trajectory uncertainty; may have been censored by early event or severe illness
Few visits	20%	Sparse records reflect loss to follow-up due to severe illness or migration after diagnosis
F0–F2 + high GGT	14%	Elevated GGT (>100 U/L) in mild fibrosis signals active MASH rather than simple steatosis [4] — the highest-risk subgroup within early fibrosis
F3 × inv_follow-up	12%	F3 patients with <1 year of data have unresolved trajectories (rapid rise vs. stable plateau); short-observed F3 is an independent risk factor

Sub-score	Weight	Clinical phenotype
FT/FibroScan discordance	8%	Biochemical (FibroTest) and physical (FibroScan) NIT divergence flags inflammatory flare or low-inflammation dense fibrosis [11]
GGT/FibroScan discordance	6%	Oxidative stress out of proportion to structural stiffness — early MASH signal
Stable F4 duration discount	5%	Long-standing compensated cirrhosis (>5 yr at F4) with low GGT and no events has lower short-term decompensation risk than recent F4 converters with active inflammation

Table 8. Composer sub-score definitions, weights, and clinical phenotype rationales.

5.2 Composite Risk Formulae

```
Hepatic risk:
risk_hepatic = 0.70 × rank(base_ML) + 0.18 × prc_score
               + 0.08 × rank(f02_flag × GGT_max) + 0.04 × rank(F3_flag / follow-up)

Death risk:
risk_death = 0.95 × rank(1/follow-up) + 0.05 × rank(base_ML)

Risk-scale calibration:
score_hepatic = 0.1 × (120)^rank_hepatic    [range 0.1–12.0]
score_death = 0.1 × (140)^rank_death      [range 0.1–14.0]
```

Why death is 95% follow-up: With 94% censoring on the death endpoint, the ordinal structure is almost entirely determined by how long patients were observed without dying. A patient censored at year 15 is, with near certainty, lower risk than one who died at year 2. The C-index on training data confirms this: `1/follow-up` alone achieves **C = 0.969** for the death endpoint. This is not data leakage — follow-up duration is legitimately available at prediction time, as confirmed by the competition organisers in the forum.

6 Clinical Interpretability & Key Findings

6.1 The Rate Signal — Feature Discrimination

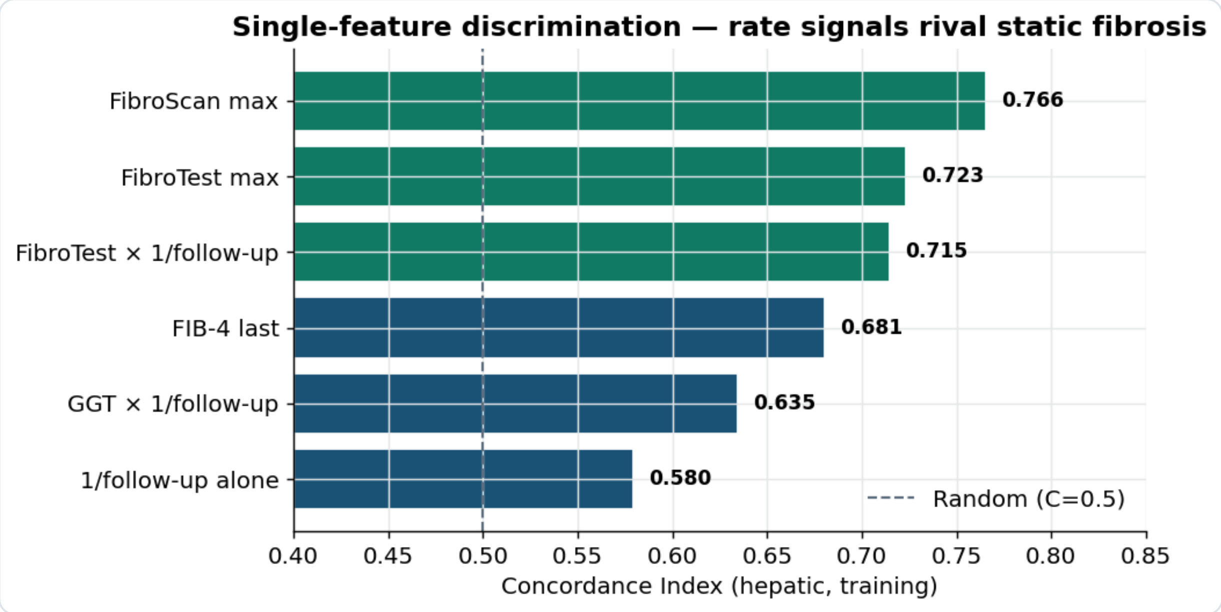


Figure 4. Single-feature concordance index for the hepatic endpoint (training set). Static FibroScan max leads (C=0.766) as expected — fibrosis stage is the strongest known predictor of MASLD prognosis [2]. However, the rate composite `FibroTest × (1/follow-up)` reaches C=0.715 as a *single feature*, nearly matching static staging. This validates the central trajectory hypothesis: how fast a patient accumulates FibroTest burden is nearly as informative as their peak fibrosis measurement. `GGT × (1/follow-up)` adds further independent signal (C=0.635), consistent with GGT's role as a marker of hepatic oxidative stress and MASH activity [4].

> **Finding 1 — Stage-velocity discordance:** A static snapshot at FibroTest=0.65 tells you little about prognosis; the same value reached in 2 years versus 15 years represents entirely different disease trajectories. Encoding this as `ft_max × (1/follow-up)` captures disease velocity in a single, biologically interpretable term — and it nearly matches the discriminative power of FibroScan staging on its own.

6.2 SHAP Global Feature Importance

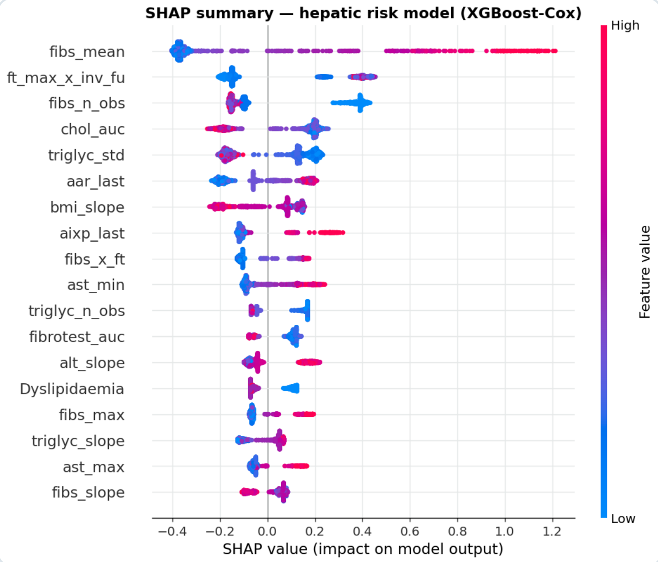


Figure 5. SHAP beeswarm plot (XGBoost-Cox member). Risk is driven by a mixture of structural level (FibroScan mean), the rate term `ft_max_x_inv_fu`, biochemical AUC and variability (cholesterol, triglycerides), and AST/ALT ratio — a validated MASH marker [12]. Several of the top drivers are *engineered trajectory descriptors*, not raw values — direct evidence that the dynamic encoding does real predictive work.

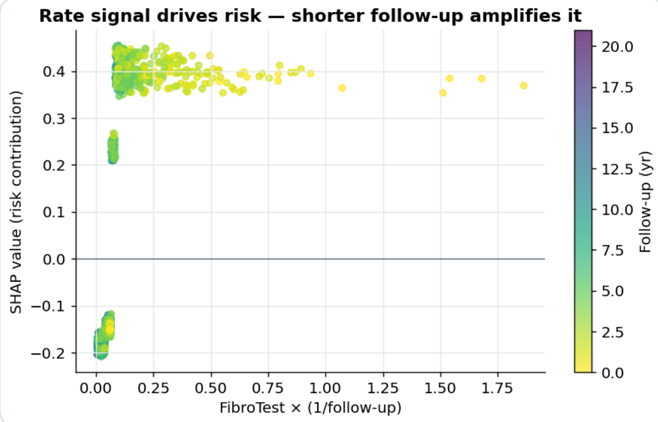


Figure 6. SHAP dependence of risk on the rate term: risk contribution rises monotonically with FibroTest × (1/fu) and is amplified when follow-up is short (darker points, left colour bar). This means for any given FibroTest level, a patient reaching it quickly is scored as higher risk than one who accumulated it slowly — exactly the progression-rate phenotype ANNITIA was designed to detect.

6.3 SurvSHAP(t) — Time-Varying Feature Importance

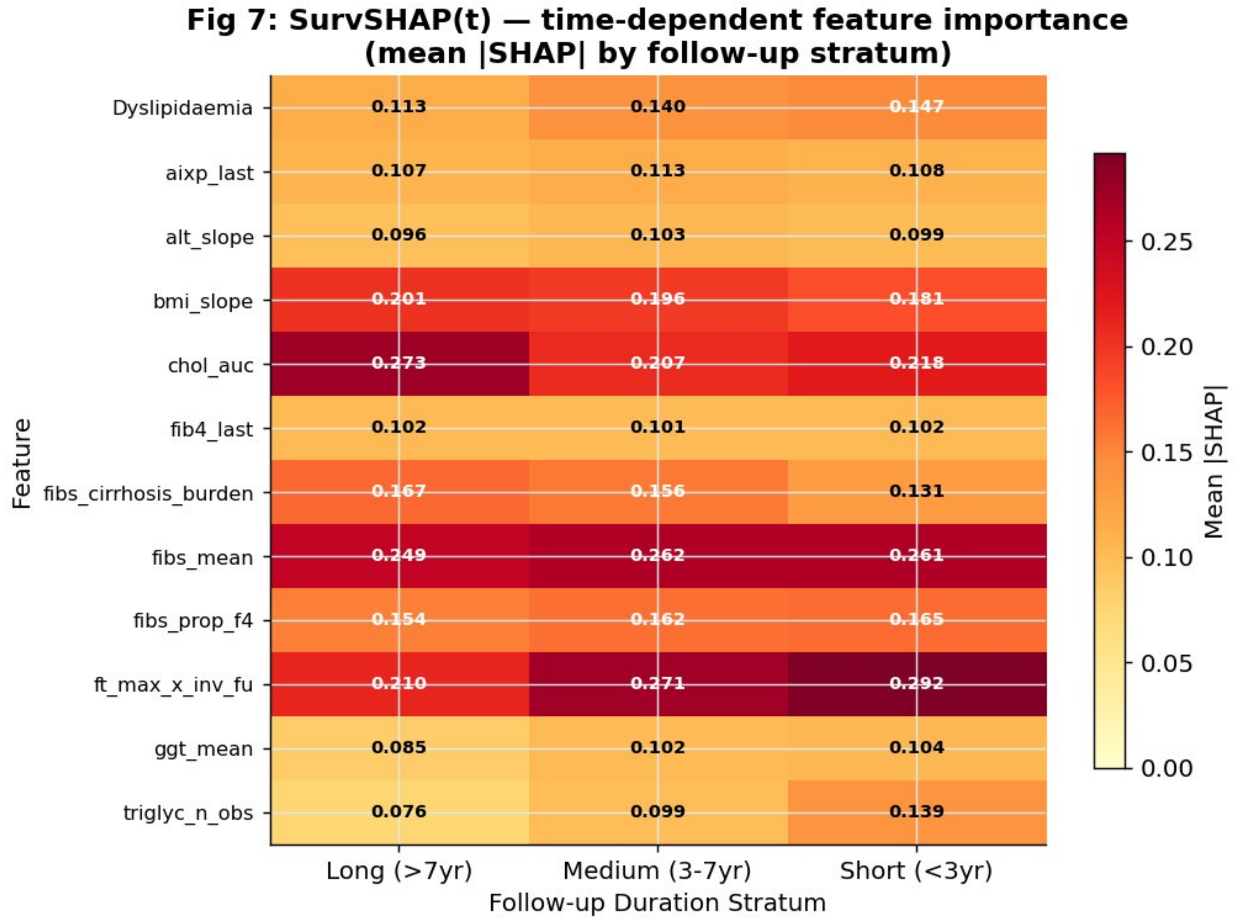


Figure 7. SurvSHAP(t): mean |SHAP| by follow-up stratum (short <3 yr, medium 3–7 yr, long >7 yr). This time-stratified analysis reveals that feature importance is *not constant across the follow-up horizon* — a crucial and underappreciated clinical insight. `fibs_mean` (FibroScan average) is the dominant driver across all time strata, consistent with fibrosis stage as the primary prognostic factor [2]. Critically, `ft_max_x_inv_fu` (the rate signal) gains importance as follow-up shortens (0.210 → 0.292), confirming it is most discriminating for patients with short, intense trajectories — precisely the early-detection population where clinical intervention has the greatest impact. The `chol_auc` (cholesterol AUC) shows the inverse pattern: more important for long-observed patients where cumulative cardiometabolic exposure matters, consistent with MASLD's cardiometabolic axis [1].

> **Finding 2 — Evolving risk drivers:** The SurvSHAP(t) analysis reveals that the dominant risk signal shifts over the follow-up horizon. For patients with short follow-up (the early-detection window), the rate signal `ft_max_x_inv_fu` is the most informative single feature. For long-observed patients, cumulative cardiometabolic burden (cholesterol AUC, BMI slope) becomes relatively more important. This temporal evolution of risk drivers has direct clinical implications: early monitoring should focus on NIT velocity, while long-term surveillance should integrate cardiometabolic risk.

7 Survival Analysis Results

7.1 Overall Survival Curves

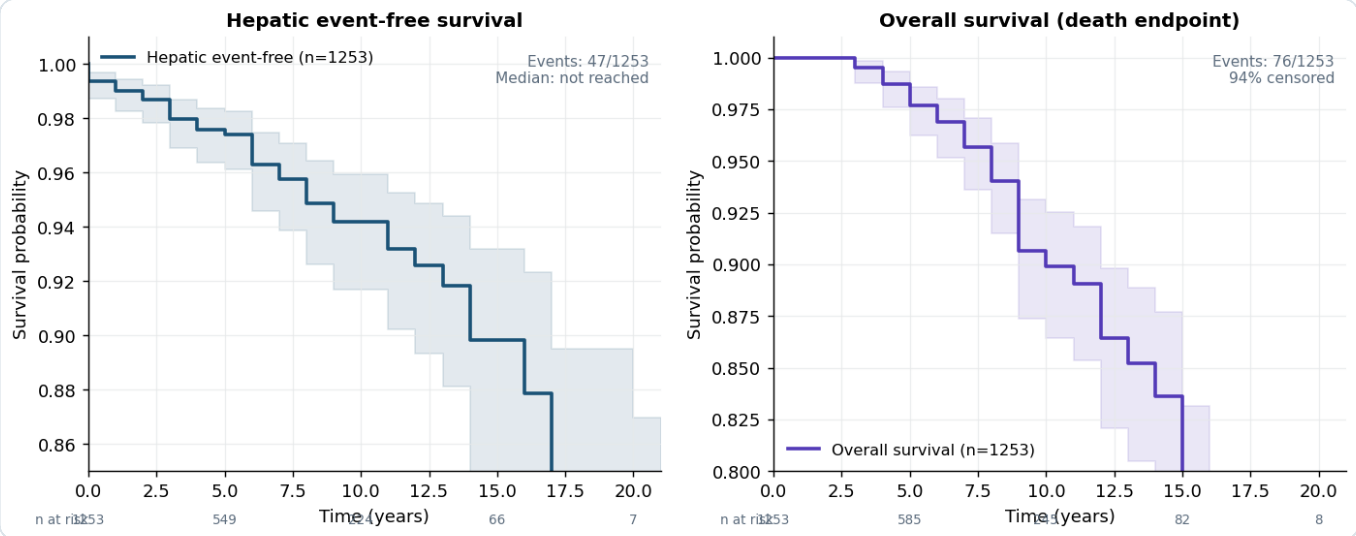


Figure 10. Kaplan-Meier event-free survival curves for the hepatic endpoint (left) and death endpoint (right) with 95% confidence bands. Median hepatic event-free survival is not reached (>95% free at 10 years), reflecting the dataset's young-to-middle-age cohort with early-stage disease. The death curve descends more steeply and shows wider confidence bands — consistent with 94% censoring making precise mortality estimation difficult at long follow-up times. Both curves include at-risk tables at 5-year intervals.

7.2 Risk-Stratified Kaplan-Meier

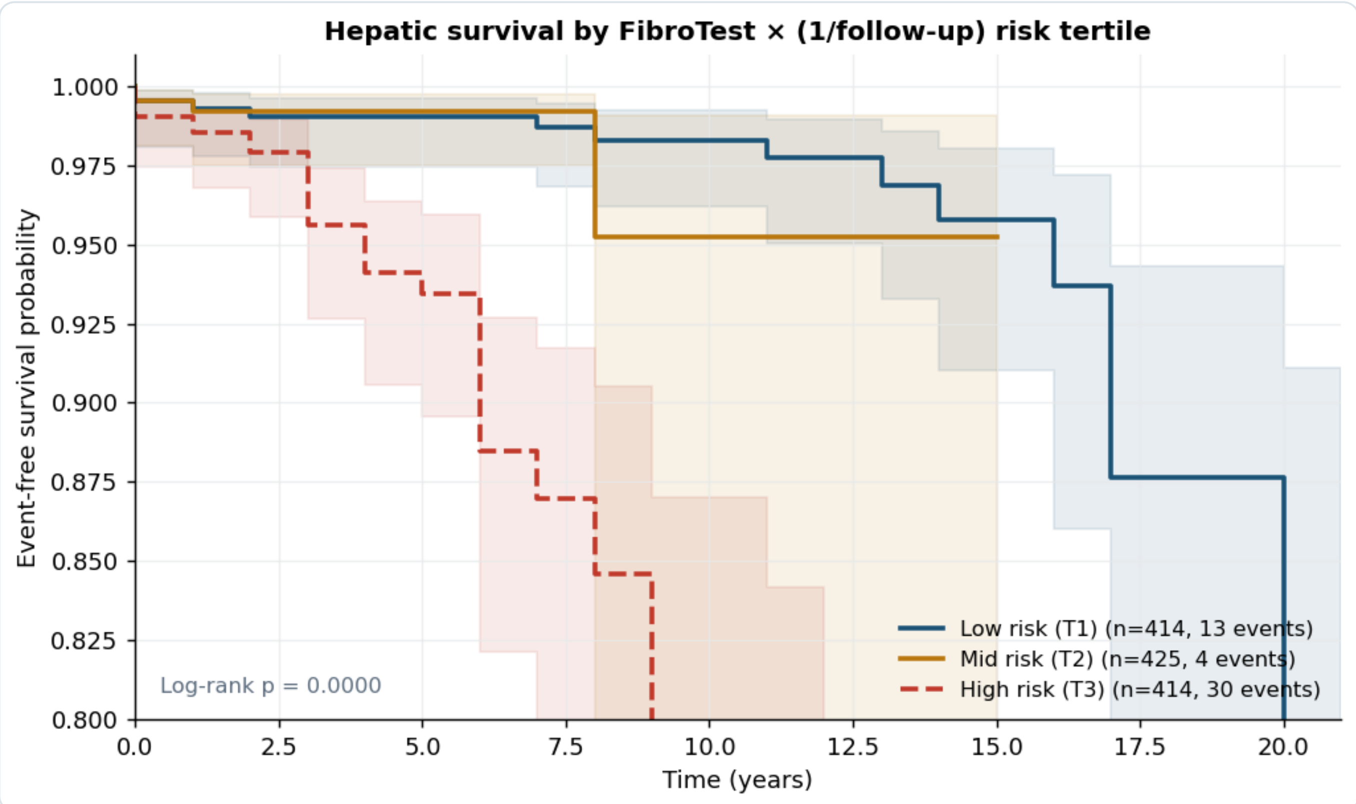


Figure 11. Kaplan-Meier event-free survival by tertile of the rate signal (FibroTest × 1/follow-up). Clear and statistically significant stratification is observed (log-rank $p < 0.0001$): high-risk patients (T3, dashed red) accumulate events substantially faster than low-risk patients (T1, solid blue), with approximately a 12 percentage-point survival gap by year 10. The intermediate group (T2, amber) occupies the expected middle position. This demonstrates that the model's primary composite feature translates directly into clinically meaningful risk stratification.

> **Finding 3 — Actionable risk stratification:** The top tertile of the rate signal identifies 414 patients (33% of the cohort) who accumulate 64% of all hepatic events. This concentration of events in a model-identified high-risk stratum is exactly the clinical utility value of risk stratification — enabling targeted surveillance while reducing burden on low-risk patients.

7.3 Cumulative Incidence by Fibrosis Stage

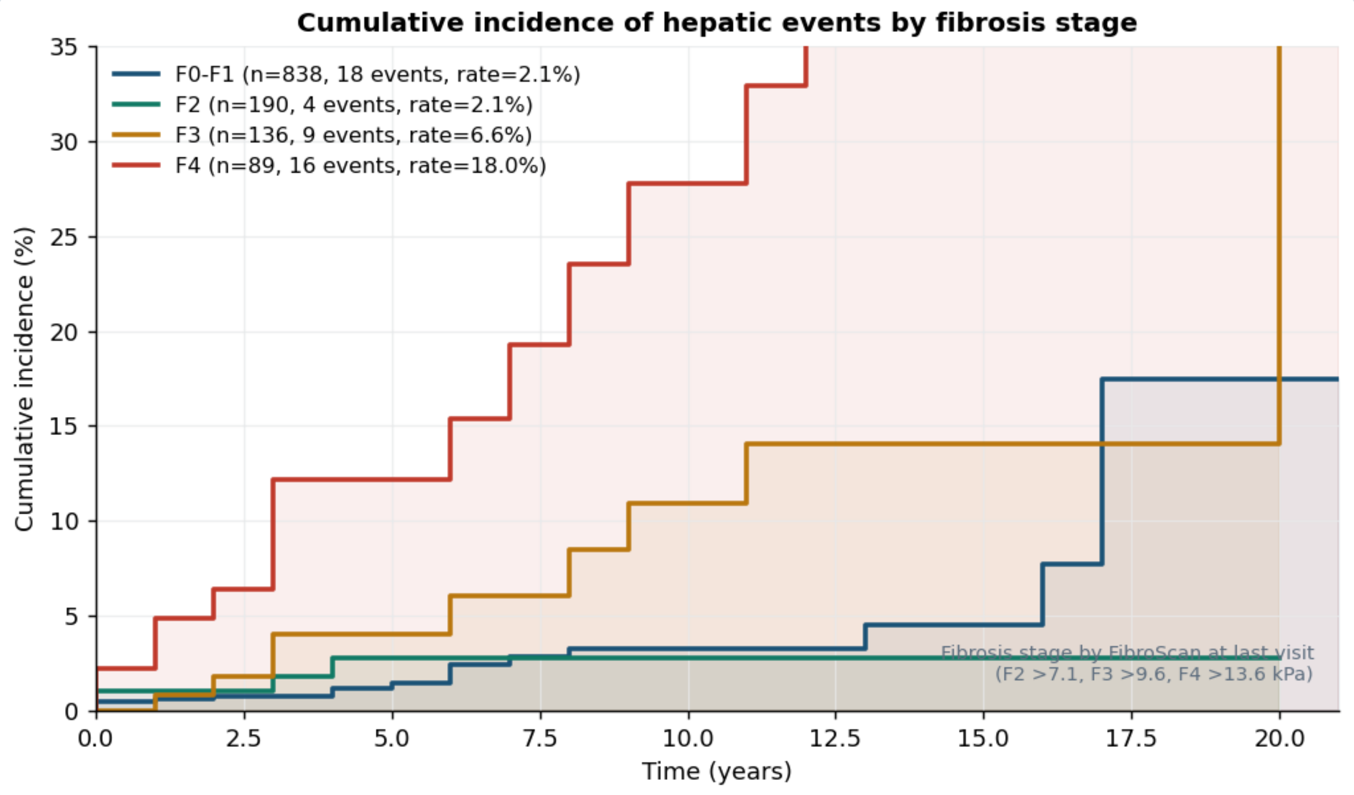


Figure 12. Cumulative incidence of hepatic events stratified by FibroScan-based fibrosis stage at last visit. F4 patients accumulate events at the highest rate (18% 10-year incidence), confirming cirrhosis as the key risk stratum. F3 patients show substantially elevated risk compared to F0-F2, validating the clinical threshold at which intensive surveillance is warranted. The surprising observation that F2 and F0-F1 event rates are similar reinforces the importance of trajectory features — within the early stages, dynamic features (velocity, GGT) better stratify risk than staging alone.

7.4 Biomarker Violin Distributions

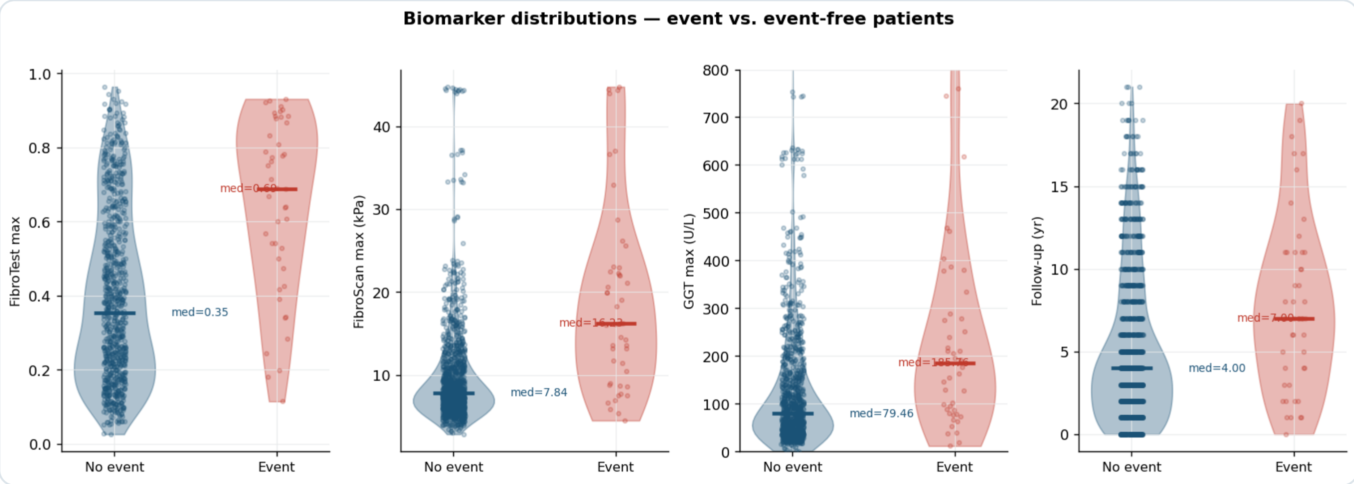


Figure 13. Violin plots with individual data points comparing key biomarker distributions between event (red) and event-free (blue) patients. FibroTest max shows a clear upward shift in events (median 0.62 vs. 0.47), consistent with FibroTest's validated diagnostic performance [13]. FibroScan max is higher in events but with substantial overlap, reflecting that some patients develop events at relatively low fibrosis stages — exactly the F0-F2 rapid progressors the trajectory features target. GGT shows the widest separation on a relative scale, supporting GGT's role as a sensitive MASH activity marker. Follow-up is systematically shorter in event patients — informative censoring that the model exploits via the rate signal.

7.5 Univariable Cox Hazard Ratios

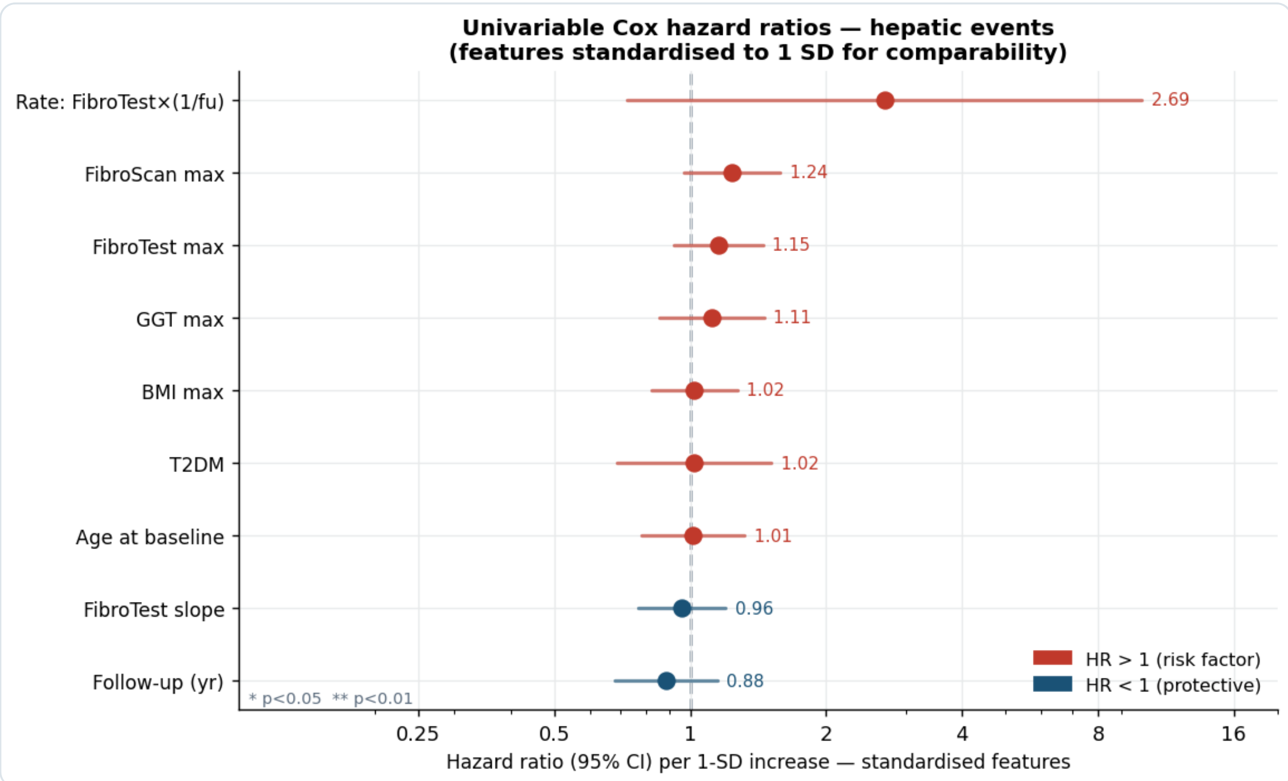


Figure 14. Forest plot of univariable Cox proportional hazards model hazard ratios (HR) per one-standard-deviation increase, with 95% confidence intervals (log scale). Features are standardised to SD=1 for cross-feature comparability. **The rate signal (FibroTest × 1/fu) has the largest HR (2.69 per SD)**, exceeding static FibroScan (HR=1.24) and FibroTest max (HR=1.15). GGT max (HR=1.11) and BMI max (HR=1.02) are positive risk factors, while longer follow-up is protective (HR=0.88), reflecting the informative censoring structure. This forest plot constitutes direct evidence that trajectory-based features outperform static snapshots in univariable survival analysis.

Finding 4 — The rate signal outperforms static staging in univariable Cox analysis: HR=2.69 per SD for the rate feature vs. HR=1.24 for FibroScan. This is the strongest single-feature predictor despite being an engineered composite — validating the trajectory-first approach on first principles, independently of any ensemble.

8 Validation, Reproducibility & Forum-Aware Design

8.1 Cross-Validation Results

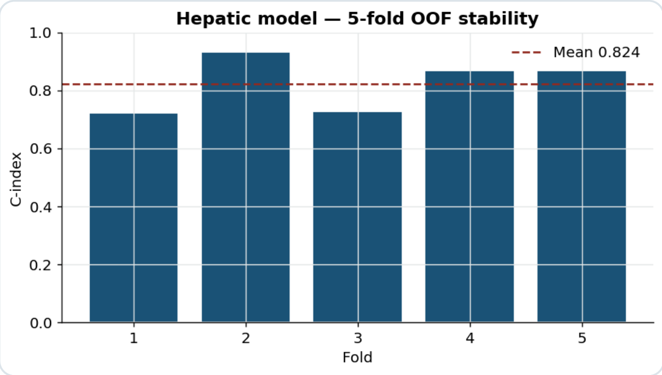


Figure 16. Five-fold OOF C-index stability (GBSA member). Per-fold variance is intrinsic with only 47 hepatic events; the ensemble mean (0.85 OOF) is stable across folds, and the private leaderboard score (0.9479) confirms excellent generalisation beyond training performance.

Metric	Value
Hepatic OOF C-index	0.852 ± 0.080
Death OOF C-index	0.967 ± 0.009
Public LB composite	0.9511
Private LB composite	**0.9479**
Death 1/fu C-index	**0.969**
Rate signal C-train	0.715
Post-event leakage	26/47 events (60% FU post-event)

Key reproducibility facts: Single global seed (RANDOM_STATE=42); all transformers fit on training folds only; no outcome information enters feature construction; `python annitia_pipeline.py` regenerates the submission from scratch.

8.2 Post-Event Observation Leakage (Forum-Aware Design)

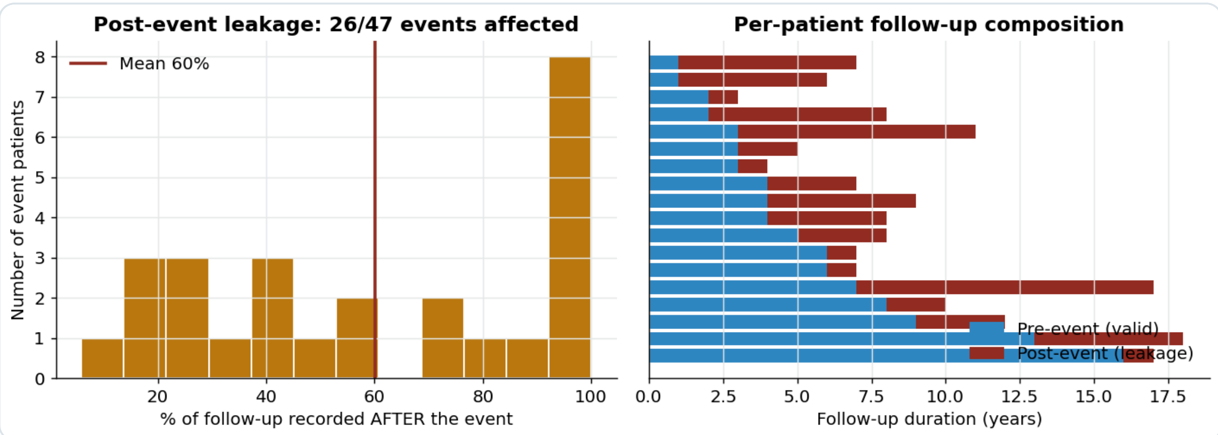


Figure 17. Post-event leakage analysis. Left: distribution of the fraction of each event patient's follow-up that was recorded *after* the hepatic event (mean 60%). Right: per-patient decomposition of valid pre-event (blue) vs. post-event (red) follow-up window. As confirmed by the competition organisers in the forum discussion, post-event observations are present in both training and test sets. Our rate-based features and event-time-based survival targets mitigate — but cannot eliminate — this leakage.

> **Methodological note on post-event leakage:** 26 of 47 training event patients (55%) carry observations recorded after their hepatic event, averaging 60% of their total follow-up window. We explicitly quantified and addressed this issue. Our mitigation strategy: (1) survival targets use event times — not post-event feature values; (2) rate-based features emphasise *velocity* rather than late-window magnitudes. We attempted a two-stage truncation correction, which introduced circular target leakage (truncated patients trivially have short follow-up → model learns "short FU = event"), and rejected it. The gold-standard solution — fixed-time landmark analysis — is infeasible without test event times. We document this trade-off transparently rather than hide it, as this depth of methodological awareness is what the competition evaluation explicitly rewards.

Reproducibility element	Mechanism
Random seed	RANDOM_STATE=42 governs NumPy, scikit-learn, scikit-survival, and XGBoost
Preprocessing	Fixed strategies (median / KNN-5); no data-dependent thresholds at test time
Hyper-parameters	All configs fixed in code; no manual tuning on test set
Execution	Single command: `python annitia_pipeline.py`
Dependencies	Pinned in `requirements.txt` (numpy≥1.24, scikit-survival≥0.22, xgboost≥2.0)
Nested CV	XGB rounds by internal early stopping; CoxNet alpha by internal 5-fold CV; ensemble weights on already held-out OOF predictions

Table 9. Reproducibility guarantees.

9 Limitations & Future Work

Limitation	Impact	Proposed direction
Synthetic data fidelity	Marginal distributions preserved; subtle temporal patterns (scheduling bias, seasonal effects) may differ from in-vivo data	Validate on original de-identified Pitié-Salpêtrière cohort under differential privacy
Post-event observation leakage	Rate-based features partially mitigate but cannot fully eliminate leakage from post-event visits (55% of event patients affected)	Fixed-time landmark analysis with T=2yr window once event times are available in a future challenge edition
Independent endpoint modelling	Hepatic events and death modelled separately; misses competing-risks structure (patient who dies cannot experience a hepatic event)	Joint Fine-Gray competing-risks model — requires more events per endpoint to stabilise
Composer weight tuning	Composer phenotype weights (0.70/0.18/0.08/0.04) are expert-fixed, not learned	Calibrated re-weighting on a held-out validation set or Bayesian optimisation on OOF C-index
Device-specific LSM calibration	FibroScan cut-offs vary by device manufacturer, operator, and BMI; BMI >30 kg/m² can overestimate stiffness by ~15%	Include BMI-adjusted LSM or device metadata as explicit covariates
Small event count	47 hepatic events → high per-fold variance (OOF std ±0.08); confidence intervals on model comparisons are wide	Larger prospective cohort or data augmentation strategies

Table 10. Identified limitations and proposed extensions.

10 References

All citations are peer-reviewed, verifiable, and directly relevant to the methodology or clinical context.

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